UMA028: MATHEMATICS FOR DATA SCIENCE

L	T	P	Cr
3	0	2	4.0

Course Objectives: To introduce the student to the concept of Probability and Statistics that plays a vital role in computing and computational intelligence. Knowledge of these topics is critical to decision making and to the analysis of data. Using concepts of probability and statistics, individuals are able to predict the likelihood of an event occurring, organize and evaluate data.

Syllabus

Mathematical Foundations of Data Sciences: Matrices, Vectors, Vector Spaces, Matrix Decomposition, Singular Value Decomposition, Eigenvalues and vectors, Sets and classes, Limit of a sequence of sets, rings, sigma-rings, sigma fields, monotone classes.

Probability: Classical, relative frequency and axiomatic definitions of probability, addition rule and conditional probability, multiplication rule, total probability, Bayes' Theorem and independence, Random variable, some common discrete and continuous distributions (Binomial, Poisson, Geometric, Rectangular, Exponential, Normal, Gamma etc.).

Bi-variate Probability Distribution: Probability distribution of functions of a random variable, Joint and marginal distributions, Conditional distributions.

Correlation and Regression: Covariance, Karl-Pearson and rank Correlation coefficients; linear regression between two variables.

Estimation: Unbiasedness, consistency, the method of moments and the method of maximum likelihood estimation, confidence intervals for parameters in one sample and two sample problems of normal populations, confidence intervals for proportions.

Hypothesis tests: Introduction to Sampling Distribution (standard normal, chi-square, T& F distributions), Theory of Estimation, Properties of an estimator, Tests for Goodness of fit: Method of maximum likelihood, Neyman-Pearson lemma (without proofs); Critical regions.
Parametric & Non-parametric tests: One sample and paired sample tests; Sign Test, Signed-rank Test, Kolmogorov Smirnov Test.
Data Processing: Regression, Dimensionality Reduction, Linear Discriminant Analysis Principal Component Analysis.

Laboratory Work

Based on the programming in MATLAB/ Python /R language of various statistical techniques. R for Data Science: Data Wrangling, Data Visualization, Programming, Python for Data Science: Python basics , NumPy, Pandas, Matplotlib.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. Compute probabilities of composite events along with an understanding of random variables and distribution functions.

2. Explain the convergence of sequence in probabilities.
3. Analyze the correlated data and fit the linear regression models.
4. Interpret the statistical inferences using principles of hypothesis tests.

Text Books:

1. Meyer P. L., Introduction to Probability and Statistical Applications, Oxford & IBH, 2007.
2. Hogg, R. V. and Craig, A.T., Introduction to Mathematical Statistics, Prentice Hall of India, 2004.
3. Ross, S.M., A First Course in Probability, 9th edition, Pearson, 2012.
4. Peng, D., R., R Programming for Data Science, Lulu.com (2012).

Reference Books:

1. Walpole, R. E., Myers, R. H., Myers, S. L. and Ye, K.,. Probability and statistics for engineers and scientists, Pearson, 2010.
2. Hestie Trevor, Tibshirani R., Friedman J., the elements of statistical learning, Springer- Verlag New York Inc., 2nd Ed., 2001.

Evaluation Scheme

Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	**70**
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	**30**